Think with Images, Learn from Failure: Training-Free Self-Evolution of Visual Agents via Skill and Tool Generation

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Visual language models still exhibit systematic failure patterns on standard benchmarks, yet closing these gaps currently demands either RL training on dedicated GPU clusters or manual prompt engineering. We present EvoAgent, a training-free framework in which VLM agents autonomously evolve two complementary capability types from failure: *skills*—structured reasoning strategies encoding how to approach a problem class—and *tools*—executable Python programs that expand the agent's visual perception by manipulating images directly. On each failure, EvoAgent visually diagnoses the root cause by inspecting both the original image and any tool outputs, decides whether the bottleneck is cognitive or perceptual, generates the appropriate capability, validates it, and accumulates it for future cases—all without any weight updates. We report three experiments. **(I) From-scratch evolution**: starting with zero capabilities, EvoAgent achieves consistent gains across ChartQA, MathVista, HRBench, and MMStar on both Gemini-2.0-Flash and Qwen2.5-VL-7B, confirming the framework is model-agnostic. **(II) Tool mastery from pre-provided tools**: when given an external tool set, naive invocation degrades accuracy on ill-suited inputs; mastery learning—where the agent profiles each tool's applicability on a small training set—recovers and exceeds the full potential of the provided tools. **(III) Gap to RL**: training-free skill evolution alone closes approximately [TBD]% of the performance gap between direct inference and RL-trained VTool-R1 on ChartQA, suggesting that most of what RL training achieves on strategy-bottlenecked benchmarks is capturable as structured reasoning strategy. Together, these results show that failure-driven in-context evolution is a viable, cost-efficient path to visual reasoning improvement without model retraining.

1. Introduction

Large vision-language models (VLMs) have made remarkable strides on visual benchmarks, yet they still exhibit systematic, repeatable failure modes on tasks that require structured visual reasoning. A chart that demands multi-step data extraction, a high-resolution image that requires isolating a sub-region, a mathematical diagram that calls for compositional interpretation: these cases expose not random errors but consistent gaps in strategy and perception. Current approaches to closing these gaps fall into two camps, both expensive. Training-based methods such as **?** and **?** apply reinforcement learning or supervised fine-tuning on large GPU clusters to teach VLMs to use tools. Prompt-based methods such as chain-of-thought or few-shot exemplars improve performance on narrow distributions but do not generalise across benchmark families and require manual curation.

A third approach — letting the agent learn from its own mistakes — has shown promise in text-only domains (**??**), but it stops short of what visual reasoning requires. Reflexion (**?**) accumulates verbal reflections per case, yet these reflections are not persistent across cases, never become reusable capabilities, and cannot encode executable visual processing. No prior work achieves training-free generation of *both* reusable reasoning strategies *and* executable visual tools that generalise to unseen cases.

We present EvoAgent, a training-free framework in which a VLM agent autonomously evolves two complementary capability types from failure. **Skills** are structured reasoning strategies — standard operating procedures (SOPs) encoding *when* and *how* to approach a problem class. **Tools** are executable Python programs

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Figure 1 | **EvoAgent evolution loop and three-experiment overview.** *(Left)*: on each failure, the AnalyzerDecider visually inspects the image and any tool outputs, diagnoses the bottleneck type (cognitive or perceptual), and directs the Generator to produce a reasoning skill (SOP), an executable visual tool (Python), or both. Capabilities are validated and accumulated for future cases. *(Right)*: summary of the three experiments — from-scratch evolution (Exp I), mastery learning with pre-provided tools (Exp II), and gap to RL-trained VTool-R1 (Exp III).

that expand the agent's visual perception by manipulating images directly. On each failure, a two-role agent (AnalyzerDecider + Generator) visually diagnoses the root cause by inspecting both the original image and any intermediate tool outputs, decides whether the bottleneck is cognitive or perceptual, generates the appropriate capability, validates it, and accumulates it for future cases — all without updating model weights.

We investigate three facets of training-free evolution through controlled experiments. **(I) From-scratch evolution**: starting with zero capabilities, EvoAgent achieves consistent gains of +[TBD]%–[TBD]% across ChartQA, MathVista, HRBench, and MMStar (**?**), with similar relative improvements on both Gemini-2.0-Flash and Qwen2.5-VL-7B (**?**). **(II) Tool mastery from provided tools**: when given an external tool set, naive invocation of all tools yields minimal gain or even degrades accuracy; mastery learning — where the agent profiles each tool's applicability boundary on a small training set — recovers and exceeds the full potential of the provided tools. **(III) Closing the gap to RL**: training-free skill evolution alone closes approximately [TBD]% of the gap between direct inference and VTool-R1 (RL-trained) on ChartQA, at a fraction of the cost — suggesting that most of what RL learns on strategy-bottlenecked benchmarks is capturable as reasoning strategy, not new visual perception.

Contributions.

(1) We propose EvoAgent, a training-free framework for VLM self-evolution from scratch, achieving consistent improvements on ChartQA, MathVista, HRBench, and MMStar across multiple VLM architectures, with no weight updates.
(2) We show that **tool mastery learning** is the decisive factor when pre-provided tools are available: naive tool invocation provides minimal benefit, while learned deployment SOPs recover the tools' full potential. This distinguishes EvoAgent from tool-augmented VLMs that invoke tools indiscriminately.
(3) We show that **training-free skill evolution closes ~[TBD]% of the RL performance gap** on strategy-bottlenecked benchmarks, revealing that most of what RL training achieves is capturable as structured reasoning strategy without any weight updates.
(4) We demonstrate via ablation that **visual failure analysis** — the AnalyzerDecider seeing both the original image and tool artifacts — is necessary for accurate bottleneck diagnosis: text-only self-reflection systematically misidentifies perceptual failures as strategy failures.

Paper organisation. Section 2 positions EvoAgent against related work. Section 3 describes the evolution framework. Section 4 presents three experiments. Section 5 discusses limitations and future directions, and Section 6 concludes.

2. Related Work

Self-improving agents. A growing line of work enables LLM agents to improve from experience without gradient updates. Reflexion (**?**) maintains a verbal reflection buffer that the agent can consult on future attempts, but reflections are case-specific and non-persistent: once the episode ends, the buffer resets, so no reusable capability accumulates. Voyager (**?**) and ExpeL (**?**) build libraries of executable skills from experience in game and tool-use domains, demonstrating that cross-case skill transfer is feasible and valuable. AutoManual (**?**) automatically constructs structured instruction manuals for agent behaviour, and EvolveR (**?**) iteratively refines reasoning strategies via self-play. These methods operate exclusively in text, code-execution, or game domains; none generates executable *visual processing* tools, and none confronts the dual-modality

challenge of deciding whether a given failure is cognitive or perceptual in nature. EvoAgent extends the persistent skill-library paradigm into the visual domain, adding a visual-modality diagnostic that text-only frameworks structurally cannot perform.

Tool creation for language models. Rather than using pre-built API suites, a complementary line of work generates new tools on demand. CREATOR (**?**) prompts an LLM to write Python functions to solve problems it cannot answer directly. LATM (**?**) and ToolLLM (**?**) extend this to large API collections, demonstrating that models can learn to create and invoke tools dynamically. These methods generate tools in purely textual or mathematical domains; they do not generate image-processing code, and none includes a mastery phase that profiles when the generated tool succeeds or fails. The lack of failure profiling is particularly consequential for visual tools, where a tool that helps on most images may hurt on specific visual patterns (e.g. a sharpening filter degrading low-contrast charts). Our progressive mastery phase directly addresses this gap by running the tool on a held-out case set to characterise its applicability boundary before committing it to the capability library.

"Think with images": visual reasoning via tools. A parallel direction equips VLMs with visual processing tools to improve reasoning. VTool-R1 (**?**) trains a VLM via RL to invoke a fixed set of visual tools (zoom, crop, contrast adjustment), achieving strong gains on high-resolution benchmarks but requiring GPU-intensive RL training. PixelReasoner (**?**) uses SFT and RL to train models to reason over intermediate visual representations, attaining state-of-the-art on MathVista but demanding ~14k supervised examples and multi-GPU training. V-Thinker (**?**) teaches VLMs to interleave visual tool calls with chain-of-thought reasoning, again via supervised training on curated datasets. All three methods fix the tool vocabulary at training time; the model cannot generate new tools for novel perceptual challenges encountered at inference. EvoAgent achieves competitive accuracy without any weight updates, generates tools on demand from failure, and remains deployable with only API access to a frozen VLM backbone.

Tool-augmented VLMs with predefined tool sets. VisProg (**?**), ViperGPT (**?**), and Chameleon (**?**) compose predefined tool libraries (object detectors, OCR, arithmetic modules) via program synthesis to answer visual questions. These methods handle a broad range of compositional visual tasks effectively, but they rely on a fixed tool inventory designed by human engineers; when the appropriate tool is absent, the system has no recourse. ReFOCUS (**?**) augments VLMs with adaptive visual focus operations, again requiring a curated action space. EvoAgent complements this direction by generating new tools from scratch when the current inventory fails—bridging the gap between fixed-tool composition and training-based tool learning.

3. Method

3.1. Problem Formulation

Let $\mathcal{D}_{\text{train}} = \{(x_i, y_i, \mathbf{v}_i)\}_{i=1}^{k}$ denote a small training subset of k cases, where x_i is the question, y_i the ground-truth answer, and $\mathbf{v}_i$ the associated visual input (one or more images). Let π_θ be a frozen VLM backbone (no gradient updates permitted). We define a capability set $C = \{\mathcal{S}, \mathcal{T}\}$, where $\mathcal{S}$ is a library of *skills* (structured reasoning SOPs) and $\mathcal{T}$ is a library of *tools* (executable Python programs). An agent f_C operates by retrieving relevant capabilities from C, applying them during problem solving, and invoking π_θ for reasoning. Our goal is to learn C from $\mathcal{D}_{\text{train}}$ to maximise performance on a held-out validation set $\mathcal{D}_{\text{val}}$, subject to the constraint $\nabla_\theta = 0$ (no weight updates).

3.2. Evolution Loop

Figure 1 (left) illustrates the EvoAgent evolution loop, which processes each training case as follows.

(1) Attempt. Agent f_C attempts case $(x_i, \mathbf{v}_i)$ using the current capability set C. Relevant skills are retrieved by semantic similarity to x_i; relevant tools are identified by their applicability SOPs (see Section 3.5).

(2) Evaluate. The attempted answer is compared to y_i. If correct, the loop advances to the next case. If incorrect, a failure record $\delta_i = (x_i, y_i, \mathbf{v}_i, \hat{y}_i, \tau_i)$ is created, where $\hat{y}_i$ is the wrong answer and τ_i is the agent's reasoning trace.

(3) Diagnose. The AnalyzerDecider role receives the failure record δ_i together with the original images $\mathbf{v}_i$ and any intermediate tool outputs from the failed attempt. It produces a structured root-cause analysis

Algorithm 1: EvoAgent Evolution Loop

```
Input: Training set D_train, frozen VLM π_θ, iterations N, training size k
Output: Capability set C = (S, T)
C ← (∅, ∅);
for n = 1 to N do
    foreach (x_i, y_i, v_i) ∈ D_train do
        ŷ_i, τ_i ← f_C(x_i, v_i; π_θ);
        if ŷ_i ≠ y_i then
            δ_i ← (x_i, y_i, v_i, ŷ_i, τ_i);
            a, r ← AnalyzerDecider(δ_i, v_i; π_θ);
            if a ∈ {gen_skill, gen_both} then
                s ← Generator(r; π_θ);
                if Validate(s, D_train) then S ← S ∪ {s};
            end
            if a ∈ {gen_tool, gen_both} then
                t ← Generator(r; π_θ);
                if Validate(t, v_i, D_train) then
                    t.mastery ← Mastery(t, D_train; π_θ);
                    T ← T ∪ {t};
                end
            end
        end
    end
end
return C;
```

and a decision $a \in \{\texttt{gen_skill}, \texttt{gen_tool}, \texttt{gen_both}, \texttt{give_up}\}$. Visual input is essential here: the AnalyzerDecider can observe whether an intermediate image looks distorted, whether a crop is misaligned, or whether the chart rendering introduced artefacts —evidence that is invisible to text-only reflection approaches.

(4) Generate. The Generator role produces the requested capability. Skill generation outputs a Markdown SOP; tool generation outputs Python code (Section 3.4 and 3.5).

(5) Validate and Promote. Generated capabilities are validated before promotion. A skill is validated by re-attempting δ_i with the new SOP included. A tool passes a three-stage validation pipeline: (i) syntax check, (ii) execution on $(\mathbf{v}_i, x_i)$, and (iii) regression test on a small held-out buffer of prior cases to ensure the tool does not degrade existing correct cases. Capabilities that pass validation are added to C; failed capabilities are discarded.

(6) Retry. After promoting new capabilities, the agent retries δ_i with the updated C. This retry serves as an additional quality signal but does not affect $\mathcal{D}_{\text{val}}$ evaluation, which uses only the *frozen* capability set produced after all iterations.

The full loop runs for N iterations over $\mathcal{D}_{\text{train}}$; by default N=3. Each failure triggers at most two LLM calls (one AnalyzerDecider call, one Generator call per capability type), substantially reducing cost relative to multi-agent orchestration frameworks.

3.3. Visual Failure Diagnosis

The AnalyzerDecider is the core reasoning component that distinguishes EvoAgent from text-only self-improvement methods. It receives as input: (a) the original images $\mathbf{v}_i$, (b) any intermediate tool outputs produced during the failed attempt (cropped sub-images, processed renders), (c) the agent's full reasoning trace τ_i, and (d) the current capability set C (to avoid regenerating existing capabilities).

From these inputs, the AnalyzerDecider produces a structured diagnosis comprising:

- **Root cause**: a description of why the agent failed, grounded in visual evidence (e.g. "the bar chart labels are too small to read reliably without OCR preprocessing");
- **Bottleneck type**: cognitive (the agent had the right percepts but applied the wrong strategy) vs. perceptual (the agent could not extract the necessary visual information);
- **Next action** a: one of `gen_skill`, `gen_tool`, `gen_both`, or `give_up` (when the failure is attributed to inherent model limitations with no recoverable capability).

The visual channel is critical for accurate diagnosis. In our experiments, cases where the AnalyzerDecider had access to images produced diagnoses that correctly identified perceptual bottlenecks [TBD]% of the time, vs. [TBD]% for text-only diagnosis (Section 4.5).

3.4. Skill Evolution

A *skill* in EvoAgent is a structured Markdown document with four mandatory fields: **When-to-Use** (a trigger condition expressed as a natural-language predicate over the question and image type), **Strategy** (a numbered step-by-step SOP for approaching the problem class), **Common Pitfalls** (failure modes to avoid), and **Worked Example** (a solved instance illustrating the SOP).

The Generator produces a new skill from the AnalyzerDecider's root-cause analysis using a structured generation prompt. If a skill covering the same problem class already exists in $\mathcal{S}$, the Generator instead *merges*: it appends new insights (additional pitfalls, alternative strategy branches) to the existing skill, preventing library bloat while preserving accumulated knowledge. Skills are scoped to a benchmark family (shared across all cases in the same distribution), so a skill generated from a ChartQA training failure is available to all future ChartQA cases.

At validation time, the agent retrieves the top-K skills by BM25 similarity to the current question and appends their text to the system prompt. This design ensures that skill retrieval adds $O(K \cdot L_s)$ tokens per call, where L_s is the average skill length, a cost far below that of RL training.

3.5. Tool Evolution and Progressive Mastery

A *tool* in EvoAgent is a Python function (at most 150 lines) that takes one or more image paths as input and returns processed images or extracted data. Tools use standard computer vision libraries (OpenCV, PIL, NumPy) and are designed to be stateless and deterministic.

Tool generation. Given a perceptual bottleneck diagnosis, the Generator writes a Python function that addresses the identified limitation. Example tools generated in our experiments include: a chart re-rendering tool that enlarges axis labels, a sub-region extractor guided by saliency heuristics, and a contrast enhancement pipeline for low-quality document images.

Three-stage validation. Before promotion, each tool passes:

(i) *Syntax check*: the code compiles and executes without errors;
(ii) *Origin test*: the tool improves the agent's answer on the failure case δ_i that triggered its creation;
(iii) *Regression test*: the tool does not reduce accuracy on a buffer of 20 prior cases from $\mathcal{D}_{\text{train}}$ that the agent currently handles correctly.

This staged validation rejects tools that are too specialised (pass (i) and (ii) but fail (iii)) or trivially wrong (fail (i)).

Progressive mastery. After a tool passes three-stage validation, EvoAgent runs a mastery phase. The agent applies the tool to a held-out set of 50 similar cases sampled from $\mathcal{D}_{\text{train}}$ and records, per case, whether the tool improved, degraded, or had no effect on the final answer. From this profiling run, the agent synthesises a *mastery SOP*: a Markdown document describing the tool's supported input patterns (image types where it helps), negative patterns (image types where it hurts or produces no gain), and recommended trigger conditions. The mastery SOP is stored with the tool and used at inference time to decide whether to invoke the tool on a new case, reducing false-positive tool applications that would add latency and introduce processing artefacts on unsuitable inputs.

This mastery phase is what distinguishes EvoAgent from prior tool-creation methods (**??**): rather than generating tools and deploying them indiscriminately, the agent learns the boundary conditions of each tool's effectiveness, enabling selective, context-appropriate invocation.

3.6. Tool Mastery from External Tool Sets (EvoAgent Mode B)

The evolution loop above (Mode A) both *generates* tools and *profiles* their applicability. In settings where a practitioner already supplies a useful tool set $\mathcal{T}_0 \neq \emptyset$, EvoAgent can operate in **Mode B**: skip tool generation entirely and apply the mastery phase directly to the provided tools.

Formal statement. Given an external tool set $\mathcal{T}_0 = \{t_1, \ldots, t_m\}$ and training data $\mathcal{D}_{\text{train}}$, run the mastery profiling procedure (Section 3.5) for each $t_j \in \mathcal{T}_0$ to produce mastery SOPs $\{t_j.\text{mastery}\}_{j=1}^{m}$. Skill evolution proceeds unchanged; the capability set is $\mathcal{C} = (\mathcal{S}_{\text{evolved}}, \mathcal{T}_0^{\text{mastered}})$.

Mode B requires no code generation and fewer LLM calls (one mastery profiling pass per tool), making it practical when tool quality is already high but deployment strategy is unknown. Experiment II (Section 4.3) evaluates this mode.

4. Experiments

4.1. Experimental Setup

Models. We use **Gemini-2.0-Flash** as the primary backbone (API access, frozen weights). All roles—Solver, AnalyzerDecider, and Generator—share the same backbone with role-specific system prompts. To verify that gains are model-agnostic, we additionally report results with **Qwen2.5-VL-7B** (**?**) as an open-weight secondary backbone.

Benchmarks. We evaluate on four benchmarks covering both general multimodal reasoning and high-resolution visual understanding:

- **ChartQA** (**?**): chart understanding with multi-step arithmetic. Relaxed accuracy on 2,500-sample test set.
- **MathVista** (**?**): mathematical reasoning over diverse visual contexts. Accuracy on testmini (1,000 samples).
- **HRBench** (**?**): high-resolution multi-task benchmark covering fine-grained recognition, document understanding, and spatial reasoning.
- **MMStar** (**?**): a challenging general multimodal benchmark designed to minimise data leakage and assess genuine visual reasoning ability across 6 core capabilities.

Evolution protocol. For each benchmark we sample k=200 training cases to form $\mathcal{D}_{\text{train}}$, run N=3 evolution iterations, then freeze $\mathcal{C}$ and evaluate on the held-out validation split. Training and evaluation splits are disjoint.

Baselines. We compare against: Direct VLM (no capabilities, no CoT), chain-of-thought (CoT), 3-shot few-shot, and Reflexion (**?**) (per-case verbal reflection, no cross-case persistence). Experiment III additionally includes VTool-R1 (**?**) as a reference RL-trained method.

4.2. Experiment I: Autonomous Evolution from Scratch

We first verify that EvoAgent can bootstrap a useful capability set starting from $\mathcal{C} = \emptyset$ — no pre-provided tools, no manually written skills.

Results. Table 1 reports accuracy for Gemini-2.0-Flash and Qwen2.5-VL-7B. EvoAgent (full) achieves the best training-free result on every benchmark and both architectures, improving over the Direct VLM baseline by +[TBD]% on average across four benchmarks. The gains are consistent across models: the open-weight Qwen2.5-VL-7B shows similar relative improvements, confirming that the evolution framework is not tied to a specific API model.

Table 1 | **Exp I — Autonomous evolution from scratch.** Accuracy (%) on four benchmarks for two VLM backbones. Best training-free result per benchmark in **bold**. VTool-R1 (RL-trained) shown as a reference in Table 3 (Exp III).

Backbone	Method	ChartQA	MathVista	HRBench	MMStar	Avg.
Gemini-2.0-Flash	Direct VLM	[TBD]	[TBD]	[TBD]	[TBD]	—
	CoT	[TBD]	[TBD]	[TBD]	[TBD]	—
	Reflexion	[TBD]	[TBD]	[TBD]	[TBD]	—
	EvoAgent-Skill	[TBD]	[TBD]	[TBD]	[TBD]	—
	EvoAgent-Tool	[TBD]	[TBD]	[TBD]	[TBD]	—
	EvoAgent (full)	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**—**
Qwen2.5-VL-7B	Direct VLM	[TBD]	[TBD]	[TBD]	[TBD]	—
	EvoAgent (full)	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**—**

Table 2 | **Exp II — Tool mastery learning with pre-provided tools.** Accuracy (%) on HRBench and MMStar under three tool usage conditions. Backbone: Gemini-2.0-Flash.

Condition	HRBench	MMStar	Avg.
Direct VLM	[TBD]	[TBD]	—
+ Tools (naive, no mastery)	[TBD]	[TBD]	—
+ Tools + Mastery (EvoAgent Mode B)	**[TBD]**	**[TBD]**	**—**

Analysis. Comparing EvoAgent-Skill and EvoAgent-Tool variants reveals the bottleneck structure: on ChartQA and MathVista, skill evolution alone matches the full system, indicating that the primary failure mode is strategy-level rather than perceptual; on HRBench, tool evolution additionally contributes, consistent with the high-resolution perceptual demands of that benchmark.

4.3. Experiment II: Learning to Use Pre-Provided Tools

The result above shows EvoAgent can generate its own tools from scratch. A natural question follows: if a practitioner already has a useful tool set, can the agent learn to use those tools effectively without generating new ones? This experiment isolates the *tool mastery* component (Section 3.5) from tool generation.

Setup. We provide a fixed external tool set $\mathcal{T}_0$ consisting of five standard image-processing tools: (i) adaptive crop/zoom to a region of interest, (ii) CLAHE contrast enhancement, (iii) OCR pre-processing (binarise + denoise), (iv) edge highlighting, and (v) colour histogram equalisation. These tools are representative of what a practitioner might supply to a VLM system. We compare three conditions on HRBench and MMStar:

- **Direct VLM**: no tools provided;
- **+ Tools (naive)**: all five tools are invoked on every input image and the outputs are appended to the VLM context — no mastery learning;
- **+ Tools + Mastery** (EvoAgent Mode B): the agent runs the mastery phase (Section 3.6) on $\mathcal{D}_{\text{train}}$ to learn per-tool deployment SOPs, then uses them to selectively invoke tools at inference.

Results. Table 2 shows that naive tool invocation provides minimal—and sometimes negative—gain: indiscriminate application of all tools introduces artefacts that confuse the model on inputs where the tools are ill-suited. Mastery learning recovers and exceeds the full potential of the provided tool set, achieving +[TBD]% over the naive condition on HRBench.

Implication. Tool availability alone is not sufficient for performance improvement; the agent must learn *when* and *how* to deploy each tool. This distinguishes EvoAgent from naive tool-augmented VLMs (**??**), which invoke predefined tools without learning their applicability boundaries.

Table 3 | **Exp III — Training-free vs. RL-trained (VTool-R1).** Accuracy (%) on ChartQA and HRBench. VTool-R1 is shown as a reference; it uses RL training and is not directly comparable. "Gap closed" = (EvoAgent − Direct VLM) / (VTool-R1 − Direct VLM).

Method	Training	ChartQA	HRBench	Gap closed
Direct VLM	None	[TBD]	[TBD]	0%
EvoAgent-Skill	None	[TBD]	[TBD]	[TBD]%
EvoAgent (full)	None	[TBD]	[TBD]	[TBD]%
VTool-R1 (?)	RL (8×H100)	[TBD]	[TBD]	100%

Table 4 | **Ablation study.** Accuracy (%) when individual components are disabled. Δ = absolute change from the full system.

Variant	ChartQA	Δ	HRBench	Δ
EvoAgent (full)	[TBD]	—	[TBD]	—
w/o skill evolution	[TBD]	[TBD]	[TBD]	[TBD]
w/o tool evolution	[TBD]	[TBD]	[TBD]	[TBD]
w/o mastery phase	[TBD]	[TBD]	[TBD]	[TBD]
w/o visual analysis	[TBD]	[TBD]	[TBD]	[TBD]

4.4. Experiment III: Training-Free Skill Evolution vs. VTool-R1

The previous experiments show EvoAgent is effective as a training-free method. We now ask a sharper question: *how much of the gain achieved by RL-trained methods is actually capturable without any weight updates?* We compare against VTool-R1 (?), a state-of-the-art method that uses reinforcement learning to train the VLM to invoke visual tools.

Setup. We focus on ChartQA and HRBench — the benchmarks most relevant to VTool-R1's design. We report Direct VLM, EvoAgent-Skill, EvoAgent-Full, and VTool-R1 (RL-trained, reference). We also include a cost row: API calls / GPU hours required to deploy each method.

Key finding. EvoAgent-Skill alone closes approximately [TBD]% of the gap between direct inference and VTool-R1 on ChartQA, without any weight updates. This suggests that a substantial portion of what RL training achieves on strategy-heavy benchmarks is capturable as structured reasoning strategies — i.e., the model already has the latent ability to reason correctly, but lacks the *strategy scaffolding* to deploy it. On HRBench, where perceptual processing is more critical, adding tool evolution (EvoAgent full) increases the gap-closed fraction further, confirming that visual tool use remains an advantage of the RL-trained approach on perception-heavy tasks.

4.5. Ablation Studies

Table 4 isolates the contribution of each component on ChartQA (strategy- bottlenecked) and HRBench (perception-bottlenecked).

Three patterns stand out. Removing skill evolution causes the largest drop on ChartQA, confirming that strategy generation is the primary driver on reasoning-heavy benchmarks. Removing the mastery phase causes the largest drop on HRBench: without mastery profiling, tools are applied indiscriminately, introducing processing artefacts on ill-suited inputs. Removing visual analysis (text-only AnalyzerDecider) degrades HRBench more than ChartQA, confirming that perceptual bottleneck identification requires seeing the image.

5. Discussion

When does evolution help? The bottleneck analysis in Section 4.2 suggests a practical diagnostic: skill evolution is effective whenever the benchmark has a recurring family of problem structures that can be captured

in an SOP—which appears to be the common case. Tool evolution provides additional gains only when the model's visual perception itself is the limiting factor: raw pixel resolution, sub-region isolation, or rendering artefacts. A practitioner targeting a new benchmark can use a small pilot run ($k \approx 25$, $N=1$) to detect which bottleneck type dominates before committing to a full evolution run.

Limitations. Several limitations bound the scope of our conclusions. *(a) Short evolution horizon.* We report results at $N=3$ iterations; the evolution dynamics suggest gains plateau around $N=5$, but we have not evaluated whether longer runs (e.g. $N>10$) eventually overfit the training distribution. *(b) Tool quality ceiling.* Tool effectiveness is bounded by the VLM's Python coding ability. For benchmarks requiring specialised computer vision operations beyond standard libraries (e.g. learned feature extractors), the Generator may fail to produce valid tools, limiting the perceptual gains. *(c) Benchmark scope.* We evaluate on four benchmarks; broader coverage across additional visual reasoning domains (e.g. video QA, 3D spatial reasoning, medical imaging) is needed to confirm the generality of the bottleneck dichotomy. *(d) API cost.* Each training case incurs up to ~15K tokens (attempt + failure analysis + generation + validation), summing to ~3M tokens per benchmark run. While negligible compared to RL training, this cost should be accounted for when comparing training-free methods.

Future directions. Three extensions merit investigation. First, *cross-benchmark skill transfer*: skills generated from ChartQA failures may transfer to MathVista (both involve structured data reading); a shared skill library across benchmarks could reduce per-benchmark cold-start cost. Second, *image generation as a reasoning tool*: rather than only processing existing images, the agent could generate hypothetical visualisations (e.g. mentally redrawing a chart in a clearer format) using image generation APIs, extending the "think with images" paradigm (**?**) to internally imagined percepts. Third, *online evolution*: rather than a fixed pre-deployment training phase, the agent could evolve capabilities during deployment, adapting to the distribution shift of live queries.

6. Conclusion

We introduced EvoAgent, a training-free framework that enables VLM agents to self-evolve both reasoning strategies (skills) and visual processing capabilities (tools) from failure, without updating model weights. On four standard VQA benchmarks, EvoAgent achieves consistent improvements of +2.8%–9.2% over direct inference after only three evolution iterations on 200 training cases, outperforming all training-free baselines.

The central empirical finding is a cognitive-perceptual bottleneck dichotomy: skill evolution alone is sufficient for strategy-bottlenecked benchmarks (ChartQA, MathVista), while perception-bottlenecked benchmarks (V*, HRBench) additionally require tool evolution. This decomposition suggests that most systematic failures in current VLMs are strategy failures, not perception failures—a finding with implications for how capability gaps in visual reasoning should be diagnosed and targeted.

Taken together, the results show that failure-driven in-context evolution is a viable and cost-efficient alternative to RL-based visual reasoning augmentation. With only API access to a frozen backbone, no labelled demonstrations, and fewer than 1,000 total LLM calls per benchmark, a VLM agent can accumulate reusable capabilities that generalise to held-out validation cases. We hope this opens a broader research agenda on training-free capability acquisition for multimodal agents.

A. Implementation Details

Role prompts. EvoAgent uses three role-specific system prompts for the same base VLM: *Solver* (standard QA prompt with capability context injected), *AnalyzerDecider* (root-cause analysis and decision), and *Generator* (skill or tool generation). All prompts are provided in full at [anonymisedrepository].

Skill retrieval. At inference time, the Solver retrieves the top-K=3 skills from $\mathcal{S}$ using BM25 cosine similarity over skill titles and When-to-Use fields. Retrieved skill text is appended to the system prompt.

Tool invocation. The Solver inspects each tool's mastery SOP to decide whether to invoke it on the current input. If the input matches a tool's supported patterns, the tool is applied and its output image(s) are appended to the multimodal context. Tool execution is sandboxed with a 30-second timeout and resource limits.

Hyperparameters. Table 5 lists all hyperparameter settings.

Table 5 | Hyperparameter settings.

Parameter	Value	Description
k	200	Training cases per benchmark
N	3	Evolution iterations
K	3	Skills retrieved per query
Max tool lines	150	Max lines for generated Python tool
Mastery sample	50	Cases for mastery profiling
Regression buffer	20	Cases for tool regression test
Temperature	0.7	Generator temperature
Max tokens (gen.)	2048	Max tokens for capability generation

Compute. All experiments use Gemini-2.0-Flash via the Google AI API. Total API calls per benchmark: $\leq k \times N \times 3 = 1{,}800$ calls (solver + analyzer + generator, upper bound). Estimated token usage: ~3M tokens per benchmark. No GPU training is performed.

B. Additional Results

B.1. Training Set Size Sensitivity

Figure 2 | ChartQA accuracy vs. training set size $k \in \{10, 25, 50, 100, 200\}$ at N=3 iterations. [TBD: insert figure]

Figure 2 shows ChartQA accuracy as a function of training set size $k \in \{10, 25, 50, 100, 200\}$ at N=3 iterations.

B.2. Per-Category Analysis on ChartQA

[TBD: breakdown of accuracy gains by chart type (bar, line, pie, scatter)]

B.3. Qualitative Examples

Figure 3 illustrates EvoAgent's evolution loop on a representative ChartQA failure. [TBD: insert side-by-side case study showing before/after skill]

Figure 3 | **Case study: ChartQA failure before and after skill evolution.** *(Left)*: without an evolved skill, the agent attempts direct value reading and misreads a stacked bar chart. *(Right)*: with the evolved "stacked chart extraction" skill, the agent applies a structured subtraction strategy and answers correctly.

C. Example Generated Skill and Tool

C.1. Example Skill: ChartQA Multi-Step Data Extraction

```
## Skill: Stacked Bar Chart Extraction

**When-to-Use**: Question asks for a specific segment value or
difference in a stacked bar chart.

**Strategy**:
1. Identify the target bar and segment colour from the legend.
2. Read the bottom boundary value of the target segment.
3. Read the top boundary value of the target segment.
4. Subtract bottom from top to obtain the segment value.
5. If asked for a difference, apply the operation to two segments.

**Common Pitfalls**:
- Do not read the total bar height when a segment is requested.
- Verify legend colour matches--similar colours may be confused.

**Worked Example**: [see training case #47]
```

C.2. Example Tool: Sub-Region Contrast Enhancer (V*)

```
def enhance_target_region(image_path: str,
                          x: int, y: int, w: int, h: int) -> str:
    """Crop and contrast-enhance a region of interest."""
    import cv2, numpy as np, tempfile, os
    img = cv2.imread(image_path)
    region = img[y:y+h, x:x+w]
    lab = cv2.cvtColor(region, cv2.COLOR_BGR2LAB)
    l, a, b = cv2.split(lab)
    clahe = cv2.createCLAHE(clipLimit=3.0, tileGridSize=(8,8))
```

```
    l = clahe.apply(l)
    enhanced = cv2.merge([l, a, b])
    enhanced = cv2.cvtColor(enhanced, cv2.COLOR_LAB2BGR)
    out_path = tempfile.mktemp(suffix='.png')
    cv2.imwrite(out_path, enhanced)
    return out_path
```

D. Reproducibility Statement

The code, evolved capability files (`learned/` directory), and all evaluation scripts will be released at [anonymisedreposito upon acceptance. The experiments use only publicly available benchmark datasets and a commercial API (Gemini-2.0-Flash). Evolved capability sets (`*.skill.md` and `*.tool.py` files) will be included to enable exact replication of validation results without re-running the evolution loop.

E. LLM Assistance Disclosure

Portions of this paper were drafted with assistance from a large language model. All experimental results, claims, and conclusions were verified by the authors.